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ORDVAC SOLUTIONS OF THE DIRICHLET PROBLEM*

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1. Introduction. The use of finite difference methods enables one to solve numerically many boundary value problems involving partial differential equations of elliptic type for which explicit analytic solutions are not available. To date these methods have primarily been used for manual computations as described for instance by Southwell,[14] Shortley and Weller,[12] Emmons,[2] Allen,[1] and Shaw[11]. Anyone who has carried out the method by hand will be impressed by the large amount of drudgery which is required, and probably feels that here is a place where the use of high speed machines is indicated. However, until recently the use of machines has been largely limited to the solution of Laplace's equation in a square, as described in Snyder and Livingston,[13] Young and Lerch,[20] and Young and Warlich,[21] although, of course, other problems have been treated.**The reason machines have not been used more is probably that, in spite of the fact that they can eliminate the drudgery of carrying out the iterations, a large amount of preparation is needed for each problem as compared with hand methods. On the other hand, manual methods are more flexible and can be quickly adapted to widely varying types of problems.

In order for the automatic procedures to have any chance of competing with manual methods, it would appear necessary to have available a large routine capable, with the insertion of constants on tape specifying the differential equation, the region, the boundary conditions, the iterative method, etc., of solving a large class of problems. The main part of the routine would be prepared once and for all, and only a small amount

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**For instance more difficult problems on weather prediction have been solved at the Institute of Advanced Study.

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of additional preparation would be required for each individual problem.

A first step toward the goal of constructing such a routine is described in this paper. An Ordvac program has been prepared which is capable of solving two difference equation analogues of the Dirichlet problem over any set of points which form a subset of the family of net points (mh, nh), $m,n = 1, 2, .. , 19$, $(h > 0)$. The detailed capabilities of the routine and its use are described in the next section. Later steps in the development would include the use of a larger number of net points, more general differential equations, curved boundaries, other types of boundary conditions, and free boundaries.

In addition to demonstrating the feasibility of such a routine, we are interested in studying empirically the rates of convergence of various iterative methods for solving the difference equations, and also in studying the accuracy of the solution of the difference equation as a solution of the differential equation.

For problems involving a large number of net points the usual iterative method, commonly referred to as the "Gauss-Seidel method,"[10,4] or as the "Liebmann method,"[6,12] converges extremely slowly. It is of considerable advantage to use a modification of this method developed independently by Frankel[3] and by the author.[18] This method involves the use of a parameter ω, known as a "relaxation factor." For a rectangular region the optimum value of ω can be computed exactly and its use results in a saving of a factor of 10 in the number of iterations. Thus instead of 20 minutes as required by the Gauss-Seidel method, the time can be cut to 2 minutes by the use of this improved method. We studied for the 5 regions shown in FIGURE 1 the variation of the required number of iterations with ω. Our aim was not only to demonstrate the large saving which the modified method permits but also to develop a method of finding the optimum ω for regions which are not rectangles and for which an exact formula is not available. As described in Section 3, it appears that a method of estimation based on areas gives fairly good results. More work remains to be done on this question, however.

In Section 4 the results obtained by using the 5-point difference equation are compared with those obtained by using the 9-point equation and also with the exact solution of the corresponding Dirichlet problem when available. Whenever it was practical to do so, more than one mesh size was used. In both of the cases involving the unit square, where the exact solution was available, the 9-point formula was found to give more accurate results. In Problem I, the case where the jump discontinuities on the boundary occurred at mesh points, the 9-point formula was 35 times as accurate, but on the other hand in Problem Ia, the case where the jump discontinuities occurred between mesh points, the 9-point solution was only about 35% more accurate than the 5-point solution. With the 5-point equation the errors were about 10 times as large for Problem Ia as

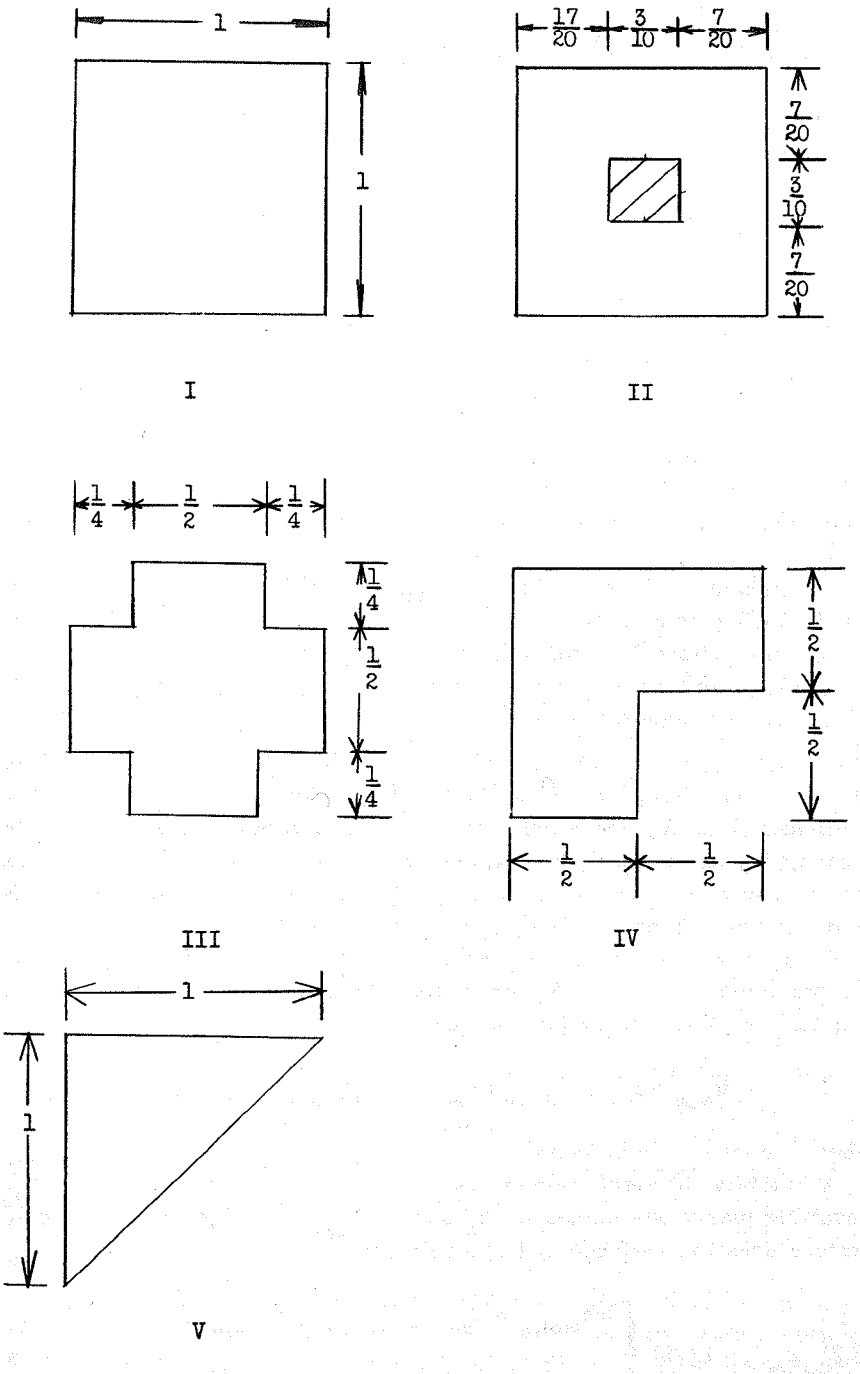


FIGURE 1. Regions Considered

they were for Problem I. With Problem I it appeared that the error for the 5-point equation varied approximately as the square of the mesh size and that as a result the use of the familiar process of extrapolation to zero grid size[8] would work fairly well. On the other hand, for Problem Ia the error was not at all proportional to the square of the mesh size. In fact, as the mesh size decreased the error changed sign. An explanation for this is given in Section 4 and a simple method is proposed which should improve the accuracy in such situations.

For problems involving regions other than the square, exact solutions of the Dirichlet problem were not available. Some indication of the error was obtained by observing the differences between the 5-point and 9-point solutions. These differences are of about the same size near corners with interior angles of 270° as they are near jump discontinuities occurring at mesh points. They are also of nearly the same size as the errors in the 5-point solution for Problem I.

The preparation of this program was begun by D.B. Gillies and was completed by Ruth M. Davis, Joanne M. Wood, and L. Ehrlich. Supplementary calculations were performed by Mrs. J.L. Vanderslice and Miriam Ezekiel.

2. Problems Solvable with the Routine. It is convenient to define the following point sets. Let L_h denote the set of all points in the plane with coordinates (mh, nh), where m and n are integers and where the positive number h is the *mesh size*. Two points (mh, nh) and $(m'h, n'h)$ of L_h are *adjacent* if $(m - m')^2 + (n - n')^2 = 1$, and are *diagonally adjacent* if $|m - m'| = |n - n'| = 1$. Let R_h be any subset of points (mh, nh) of L_h such that $1 \leqq m,n \leqq 19$, and let S_h be the subset of L_h which is disjoint from R_h and which consists of points each adjacent to some point of R_h. Also, let S_h' be disjoint from R_h and consist of points which either belong to S_h or are diagonally adjacent to some point of R_h. Evidently if (mh, nh) belong to S_h or S_h', then $0 \leqq m,n \leqq 20$.

We consider the problem of finding $u(x, y)$ defined on $R_h + S_h$, assuming prescribed values on S_h and satisfying in R_h the 5-point difference equation analogue of Laplace's equation

$$u_{m,n} = 1/4\,(u_{m+1,n} + u_{m-1,n} + u_{m,n+1} + u_{m,n-1}), \tag{2.1}$$

where we let $u_{m,n} = u(mh,nh)$.

A slightly different problem is to find $u(x,y)$ defined on $R_h + S_h'$, assuming prescribed values on S_h' and satisfying in R_h the 9-point difference equation analogue of Laplace's equation

$$u_{m,n} = 1/20 \left\{ \begin{array}{l} 4(u_{m+1,n} + u_{m-1,n} + u_{m,n+1} + u_{m,n-1}) \\ + u_{m+1,n+1} + u_{m+1,n-1} + u_{m-1,n+1} + u_{m-1,n-1} \end{array} \right\} \tag{2.2}$$

Each of the above problems may be considered as a difference equation analogue of a Dirichlet problem associated with a region R with boundary S such that $R_h \subset R$ and $S_h \subset S$ in the case of the 5-point formula, and $S_h' \subset S$ in the case of the 9-point formula. Although R and S are by no means unique, nevertheless, in the present discussion we shall, unless otherwise stated, assume that S consists of lines each containing at least one point of S_h, or S_h', and making angles of 0°, 45°, 90°, or 135° with the x axis.

To solve either of the difference equations we choose a tentative function $u'_{m,n}$ which vanishes identically in R_h and successively modify this function. In the Gauss-Seidel method we choose an ordering of the points of R_h, and, taking the points in this order, replace the value of $u'_{m,n}$ at each point by an "improved" value found by substituting the approximate values in the right member of equation (2.1) or (2.2), as appropriate. In all cases the improved values are used as soon as available, the old values being discarded as the new values are computed. Having modified $u'_{m,n}$ at all points of R_h, we start over using the same ordering as before and repeating as often as necessary until $\sigma_n \leqq \epsilon$, where ϵ is a prescribed tolerance and σ_n is the maximum of the absolute values of the residuals after n iterations. A residual is defined as the difference between the right and left members of equations (2.1) or (2.2) as appropriate.

The rapidity of convergence can be greatly increased by the introduction of a parameter ω, satisfying the condition $1 \leqq \omega < 2$. The modified method is the same as the Gauss-Seidel method except that at each step one takes as the improved value the sum of the old value and ω times the difference between the value obtained by using equations (2.1) or (2.2) as appropriate, and the old value. With $\omega = 1$ we have, of course, the Gauss-Seidel method. Methods for choosing ω are discussed in the next section.

As the routine is now set up, we are restricted to orderings of the following type. We assign numbers 1, 2, 3.... to the points in FIGURE 2, starting with 1 at the point $(h, 19h)$ and following the order of reading the words on a page, i.e. from left to right and down, thus ending at the point $(19h, h)$ which is numbered 397. To specify the ordering one selects a positive integer p and improves the values at points labeled $k_1, k_2, \ldots$ where $k_1 = 1$, and $k_q \equiv 1 + qp \pmod{397}$.* At each step the machine examines the point in question to determine whether or not it belongs to R_h. If the point does belong to R_h, then the improved value is computed as described above, otherwise the machine proceeds at once to the next point. In most of our problems we chose $p = 2$.

Using the routine one can solve equations (2.1) or (2.2) for any region of the type described above, and with any (suitably scaled) boundary values. After reading in the main part of the routine, which is used

*It is easy to show that since 397 is a prime, the sequence $\{k_1\}$ is periodic with 397 as the smallest period.

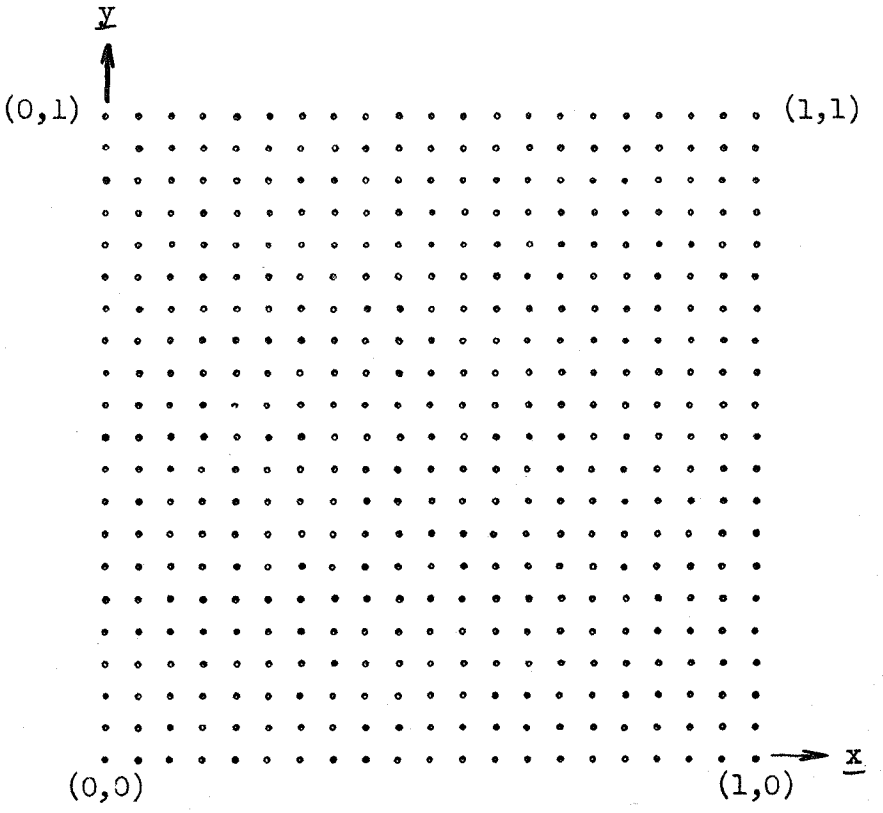


FIGURE 2. Available Net Points

for all problems of the class, one then reads in an auxiliary tape with numbers specifying the boundary S_h or S_h', the boundary values, the difference equation [(2.1) or (2.2)], and the constants t, ϵ, p, and ω. After every t iterations the machine prints n, the total number of iterations which have been performed, as well as σ_n and σ_n/σ_{n-1}. When the iterative process has converged, the final values of these quantities are printed together with all values of the solution in R_h.

The routine has been used to solve Dirichlet problems involving the five regions shown in FIGURE 1. The exact boundary values are given in Section 4 where we discuss the accuracy of the difference equation analogues. Approximately 2 seconds per complete iteration were required to compute the improved values. The same amount of time per iteration was also used to compute the residuals, although this time could have been eliminated if had we been satisfied with a slightly less accurate convergence criterion, namely that the maximum difference between the old and the improved value for a complete iteration should be less than a prescribed tolerance. This criterion was used in Young and Lerch.[20] However, since we wished to determine as accurately as possible the

effect of the choice of ω on the rate of convergence, it seemed that the additional computation time would be justified. It is a very simple matter to modify the program to use the faster process whenever desired.

3. Rates of Convergence. TABLE I shows the observed number of iterations, N, with various values of ω for each problem and for both the 5-point and 9-point formulas. It should be remarked that for a given problem the final solution of the difference equation was, of course, the same for all values of ω. Graphs of N vs. ω are given in FIGURES 3 and 4. The curves in FIGURE 3 are based only on results for the 5-point formula.

Before discussing these results we shall first describe a little of the underlying theory. Let λ denote the spectral norm (i.e. the maximum of the absolute values of the eigenvalues) of the linear transformation, T, which represents the Gauss-Seidel method. This linear transformation is well defined since with one complete iteration we transform, in a linear manner, the vector whose components are the errors in the approximate

TABLE I

Observed Values of N for Various Values of ω.

	Problem I		Problem II		Problem III		Problem IV		Problem V	
ω	5pt	9pt	5pt	9pt	5pt	9pt	5pt	9pt	5pt	9pt
1.00	704	598*	212	179	477	398*	374	315*	298	256*
1.50	..	..	..	49	..	..	..	..	..	..
1.51	..	..	..	..	..	..	..	..	..	..
1.52	..	..	..	43	..	..	..	..	..	..
1.53	..	..	48	40	..	..	..	..	..	..
1.54	..	..	44	38	..	..	..	..	..	..
1.55	..	..	37	39	..	..	..	..	73	..
1.56	..	..	37	40	..	..	..	..	..	..
1.57	..	..	39	..	..	..	..	..	..	54
1.58	..	..	..	..	..	..	..	..	..	50
1.59	..	..	..	..	..	..	..	..	..	46
1.60	..	..	..	..	..	..	..	..	51	44
1.61	..	..	..	..	..	..	..	59	45	..
1.62	..	..	..	..	..	..	..	54	45	47
1.63	..	..	..	..	..	..	..	51	..	..
1.64	..	..	..	..	..	71	54	51	..	..
1.65	..	..	..	..	79	66	50	53	50	..
1.66	..	..	..	..	..	60	52	..	..	..
1.67	..	..	..	..	..	57	54	..	..	..
1.68	..	..	..	..	58	59	..	..	..	..
1.69	..	..	..	..	59	..	..	..	..	..
1.70	..	82	..	..	62	64	60	64	..	..
1.71	94	76	..	..	..	..	..	..	..	..
1.72	84	70	..	..	..	..	..	..	..	..
1.73	71	72	68	..	68	73	68	71	68	..
1.74	70	75	..	..	..	..	..	..	..	..
1.75	74	78	..	..	..	..	..	..	..	..
1.7325	68	..	..	..	..	..	..	..	..	..
1.675	..	..	..	..	60	..	..	..	..	..

*Estimate based on 200 iterations.

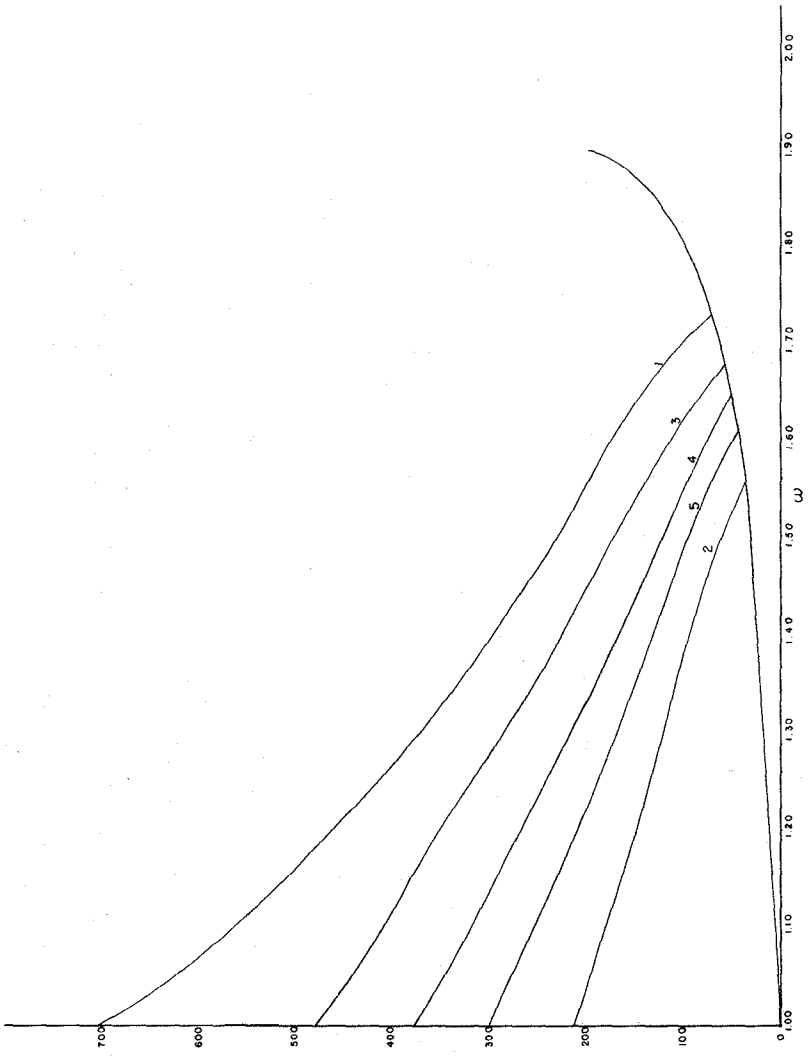


FIGURE 3. Variation of N with ω.

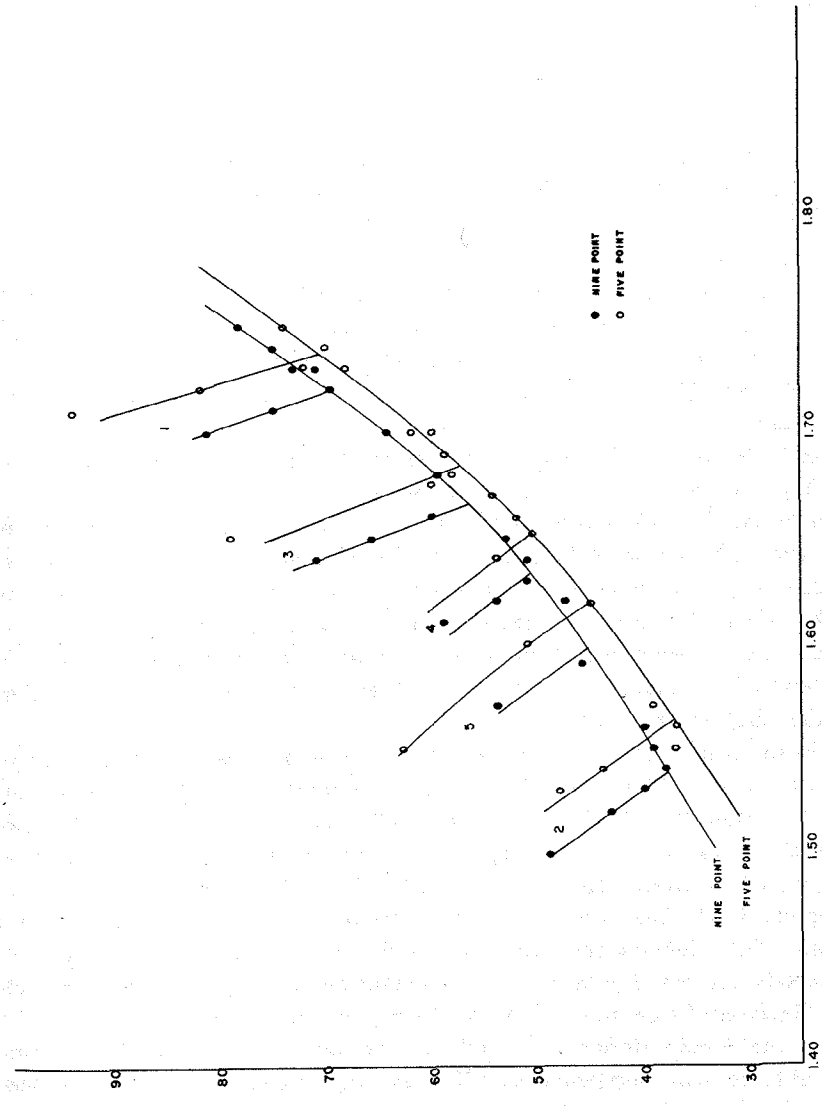


FIGURE 4. Variation of N with ω: (large scale).

solution at the points of R_h into the corresponding error vector for the improved values. It was shown by Frankel[3] for the case of Laplace's equation for the rectangle, and by the author[18] for self-adjoint elliptic difference equations and for the more general class of regions described above, that ω_b, the "best" value of ω, is given by

$$\omega_b = 1 + \frac{\lambda}{(1 + \sqrt{1 - \lambda})^2} \tag{3.1}$$

and that the rate of convergence of the method using this value of ω_b is $R(\omega_b) = -\log(\omega_b - 1)$. The value ω_b is best in the sense that the spectral norm of the linear transformation corresponding to ω_b is less than that corresponding to any other value of ω. The rate of convergence of a linear transformation is the negative logarithm of the spectral norm of the transformation and is approximately, as $\lambda \to 1-$, inversely proportional to N. In Young[18] it is shown that as $\lambda \to 1-$, we have $R(\omega_b) \sim 2\sqrt{R(1)}$, from which it follows that if $R(1)$ is very small, as it is in the cases which we are considering, then $R(\omega_b)$ is many times greater. Thus the use of ω_b enables us to obtain convergence much more quickly than is possible with the Gauss-Seidel method.

In Young,[18] it is shown that if ω is slightly larger than ω_b, then the increase in N is relatively small, but if ω is less than ω_b, then there is a much larger increase in N. Moreover, if ω is larger than ω_b, then the number of iterations is independent of ω_b – that is the use of an ω which is too large should result in nearly the same number of iterations regardless of whether the ω_b for the problem in question is much smaller or only slightly smaller than ω.

These statements seem to be well borne out by the results (see TABLE I and FIGURE 4) not only for the 5-point but also for the 9-point formula, even though the theory has only been proved for the 5-point formula. This is not surprising since although the matrix of the difference equation arising from the use of the 9-point formula does not have "Property (A)" (see Young[18]), which property was used in the proofs of most of the results concerning the variation of N with ω, nevertheless the matrix for the 9-point difference equation is, roughly speaking, not very "different" from the matrix for the 5-point difference equation.

For the 5-point formula the actual factor of reduction of the number of iterations was somewhat smaller than the theoretical value, but the gain was still appreciable, amounting to more than 10 for Problem I. For the 9-point formula the factor of reduction was less in each case than for the 5-point formula.

We now consider the problem of estimating ω_b, the best value of ω. By (3.1) the problem of estimating ω_b for the 5-point formula is equivalent to the problem of estimating λ, defined above. For any rectangle it

is easy to compute λ exactly, see for instance Young.[18] For the unit square with mesh size h we have

$$\lambda = \cos^2 \pi h. \tag{3.2}$$

Computing ω_b from (3.1) with this value of λ gave 1.73, and this was found to be well within .01 of the value of ω which yielded the fewest iterations.

Of course one may rightly ask how ω_b should be determined for regions which are not rectangles. Because of the relatively small adverse effect on the rate of convergence when an ω is used which is slightly larger then ω_b, we seek estimates of ω which err, if at all, on the high side. We shall discuss several methods and show how they apply for the 5 problems which were solved.

When $\omega = 1$ it is observed that the ratios σ_{n+1}/σ_n approach a constant as n increases. (See Section 2 for definition of σ_n.) According to theory this constant should be λ, and indeed we found with Problem I that after 100 iterations the theoretical value of λ given by (3.2) and the value of σ_{n+1}/σ_n agreed to 4 decimal places. Moreover, it was found that in each case, if the computation of ω_b was based on this observed value of λ, then the result was within .01 of the value of ω_b which gave the fewest iterations for the 5-point formula and within .02 for the 9-point formula. Although the differences are sometimes in the wrong direction, nevertheless, they are so small that at most 4 extra iterations were required with the 5-point formula and at most 6 extra iterations were required with the 9-point formula.

Evidently this empirical method for choosing ω gives very satisfactory results. However, about 100 iterations were required to find λ – more than would be required to solve the problem using ω_b. This effort would not appear to be justified unless one had many cases to solve with the same region. Unfortunately, as shown in TABLE II, one does not obtain a very good estimate of ω_b from running only 20 iterations with $\omega = 1$.

An alternative method for choosing ω_b for a given region is to use the ω_b for a circumscribing rectangle. It can be shown that an ω_b so chosen will be at least as large as the correct value. On the other hand it was observed that with Problem II, 84% more iterations were required using the value of ω_b for the circumscribing unit square, than would have been required using the truly best value of ω. For the other problems the increase was 33% or less.

Another method is to use the ω_b for an approximating rectangle with the same area. Actually we used a square of the same area. Since it can be shown that the ω_b for a square is larger than the ω_b for any other rectangle of the same area, using a square tends to increase ω_b. As

shown in TABLE II, the values of ω_b thus computed were always larger than the best value of ω, and it would be interesting to try to prove this in general. Also, the maximum number of extra iterations required was less than 15% of the smallest possible value except for Problem II where the increase was 50%.

For rectangles it can be shown that with a given area the larger the perimeter the smaller the number of iterations. Assuming this to be true for other regions as well, one could explain why the number of iterations with Problem II was less than for Problems III and IV. For, all three

TABLE II

Comparison of Various Methods for Estimating the Best Value of ω.

	Problem I		Problem II		Problem III		Problem IV		Problem V	
	5pt	9pt	5pt	9pt	5pt	9pt	5pt	9pt	5pt	9pt
ω_1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
$N(\omega_1)$	704	598*	212	179	477	398*	374	315*	298	256*
ω_c	1.73	1.71	...	...	...	...	...	...	...	...
$N(\omega_c)$	71	76	...	...	...	...	...	...	...	...
ω_o	1.7325	1.72	1.55	1.54	1.68	1.67	1.65	1.63	1.61	1.60
$N(\omega_o)$	68	70	37	38	58	57	50	51	45	44
ω_E	1.73	1.71	1.55	1.52	1.675	1.65	1.64	1.62	1.61	1.58
$N(\omega_E)$	71	76	37	43	60	66	54	54	45	50
ω_R	1.73	1.71	1.73	1.71	1.73	1.71	1.73	1.71	1.73	1.71
$N(\omega_R)$	71	76	68	67	68	67	68	67	68	67
ω_{20}	1.67	...	1.55	...	1.63	...	1.63	...	1.59	...
$N(\omega_{20})$	127*	...	37	...	93*	...	59*	...	56*	...
ω_A	1.73	...	1.69	...	1.69	...	1.69	...	1.64	...
$N(\omega_A)$	71	...	57*	...	59	...	57*	...	48*	...

$N(\omega)$ = No. of iterations necessary to reduce σ to 2^{-32} using ω.
$\omega_1 = 1$.
ω_c: Computed from exact formula for the 5-point formula. For the 9-point formula based on an exact value of μ, the spectral norm for Jacobi's method, and from (3.1) with λ replaced by μ^2.
ω_o: Observed to give the fewest iterations.
ω_E: Empirical based on observed values of σ_n/σ_{n-1} for $n = 100$.
ω_R: The best ω for the circumscribing square.
ω_{20}: Empirical based on observed values of σ_n/σ_{n-1} for $n = 20$.
ω_A: The best ω for a square of the same area.
*: Estimated

problems involve regions of the same area, but the perimeter of the region associated with Problem II is considerably larger than the perimeter of either of the other regions.

It is tempting to try to infer the rapidity of convergence for a given ω by using that ω for a few iterations, observing the values of σ_{n+1}/σ_n and modifying ω in some way, perhaps automatically by the machine. However, although the sequence $\{\sigma_{n+1}/\sigma_n\}$ approaches a constant value for $\omega = 1$ since all eigenvalues of the associated linear transformation are real, unfortunately when $\omega > 1$ these eigenvalues are often complex and the values of σ_{n+1}/σ_n oscillate to such an extent that even by taking averages it is hard to compare the effectiveness of various values of ω.

TABLE II summarizes the results obtained from using the various methods for estimating ω_b. On the basis of these results it seems that the area method for estimating ω_b is best if only a few cases with the same region are to be solved. It would be desirable to modify slightly the ω after each case to attempt to get faster convergence. On the other hand, if a great many cases are desired for a given region, one might be justified in determining λ accurately by running one case for many iterations with $\omega = 1$. Another possibility would be to try to infer the correct value of λ by running with $\omega = 1$ and a coarse mesh. This procedure would certainly reduce the time required to find λ although the value found would presumably be less accurate. We have not yet tested the effectiveness of this procedure.

Some studies were carried out to determine the effect on the convergence rate of different boundary values and various orderings of the points of R_h. On the basis of fragmentary evidence it appears that the effect of these factors is small. The constant p, defined in the previous section, was usually taken as 2.

4. Accuracy of Difference Equation Solutions. Since finite difference methods are frequently used, it seems somewhat surprising that so little is known about the accuracy of the solution of the difference equation as a solution of the related Dirichlet problem. Gerschgorin[5] has obtained an error bound which involves bounds on the fourth partial derivatives of the solution of the Dirichlet problem in the closed region. However, this result is of limited applicability since the solution of the Dirichlet problem itself is seldom known, to say nothing of the derivatives (although the desired derivatives can sometimes be estimated fairly accurately by computing suitable difference quotients). On the other hand, if in a given case it were known that the fourth partial derivatives were bounded in the closed region, then one could conclude from Gerschgorin's result that the error would vanish at least as fast as the square of the mesh size, h. The assumption that the error is exactly proportional to h^2 leads one to Richardson's process of extrapolation to zero grid size.[8] Thus, if $u_h(x,y)$ and $u_{h/2}(x,y)$ represent solutions of

the difference equation with mesh sizes h and $h/2$ respectively, then the extrapolation process would give the result

$$u(x,y) \sim (4/3)\, u_{h/2}(x,y) - (1/3)\, u_h(x,y).$$

Similarly, one can show that if the sixth partial derivatives of $u(x,y)$ are bounded in the closed region, then for the 9-point formula the error tends to zero like h^6 and, assuming the error to be exactly proportional to h^6, one would obtain by the extrapolation process

$$u(x,y) \sim (64/63)\, u_{h/2}(x,y) - (1/63)\, u_h(x,y).$$

Recent results on the accuracy problem have been obtained by Rosenbloom,[9] Wasow,[17] and Walsh and Young.[15] However, none of these papers gives a satisfactory method for determining the error in the general case. Indeed, Rosenbloom's error bound, like Gerschgorin's, involves quantities which depend on the solution of the differential equation itself, while the results of Wasow and of Walsh and Young apply only to rectangles.

It is our purpose in this section to discuss the accuracy question in the light of our numerical results. We give below the boundary values which were prescribed for each of the problems considered. The regions are shown in FIGURE 1.

Region	*Problem*	*Boundary Values*
I	I	$u(x,1)=1$, $(0<x<1)$; $u(0,1)=u(1,1)=1/2$; $u=0$ elsewhere on S.
I	Ia	$u(x,1)=1$, $(1/3<x<2/3)$; $u=0$ elsewhere on S.
II	II	$u=1$ on inner boundary; $u=0$ on outer boundary.
III	III	$u(0,y)=u(1,y)=1$, $(1/4<y<3/4)$, $u(x,0)=u(x,1)=1$, $(1/4<x<1)$; $u(1/4,0)=u(3/4,0)=u(1/4,1)=u(3/4,1)=u(0,1/4)=u(0,3/4)=u(1,1/4)=u(1,3/4)=1/2$; $u=0$ elsewhere on S.
IV	IV	$u(x,1)=0$, $(0<x<1)$, $u(0,1)=u(1,1)=1/2$; $u=0$ elsewhere on S.
V	V	$u(x,1)=0$, $(0<x<1)$; $u(0,1)=u(1,1)=1/2$; $u=0$ elsewhere on S.

We remark that for Problem V the set S_h' does not belong to S. Hence for the 9-point formula we assume that the slanting side of S is replaced by a broken line consisting of segments of length h, each containing a point of S_h and a point of S_h' and making, alternately, angles of 0° and 90° with the x axis.

For purposes of studying the accuracy of the difference equation solutions we will not consider Problem V. The solution of the Dirichlet problem and the 5-point difference equation analogue can be obtained in an obvious way by adding the solutions of two problems similar to Problem I, because of the linearity of the differential and difference equations. Hence the errors for Problem V could be computed from those of Problem I. On the other hand, for the 9-point equation the special treatment necessary to handle the slanting side is likely to cause unusual behavior. The only reason for considering the problem at all was to study the number of iterations required for convergence.

TABLE III shows the exact solution for Problem I, together with the errors E_k and F_k where

$$E_k = u_{1/k} - u$$

$$F_k = v_{1/k} - u.$$

In general, u_h and v_h denote respectively the solution of the 5-point and 9-point difference equations with mesh size h. The exact solution of the Dirichlet problem, denoted by u, was computed by means of a conformal mapping of the square into the unit circle* followed by the use of Poisson's formula.

The numbers corresponding to each point are arranged as indicated**

E_5 /(1/16)E_5 (when available)

E_{10}/(1/4) E_{10} (when available)

E_{20}

u

F_{20}

F_{10}/(1/64)F_{10} (when available)

F_5 /(1/4096)F_5 (when available)

*The elliptic functions involved in the mapping function were evaluated by means of the tables by G. W. Spenceley and R. M. Spenceley, *Smithsonian elliptic functions tables*, vol. 109, Smithsonian Institution, Washington, 1947.

**For Problem I, the values of u, E_5, E_{10} and E_{20} are given in Young and Lerch[20] to one less significant digit.

TABLE III

Numerical Results for Problem I.

y	x	.05	.10	.20	.30	.40	.50
.95		-9	-7 197	-2 567	-960	-507	-397
		497 265	699 319	832 651	876 814	894 658	899 649
		-1	240	19	2	1	9
.90	Not		-135/-34	-7472/-1868	-5062/-1265	-3373/-843	-2870/-718
			-34	-2 371	-1 371	-854	-715
	Computed		489 060	682 262	759 422	792 315	801 690
			0	15	3	1	0
			0/0	241/4	41/1	12/0	7/0
.80	Not		---	-1776/-112	---	-11338/-709	---
			6960/1740	-465/-116	-3142/-786	-3449/-862	-3395/-849
			2 244	-117	-877	-913	-882
	Computed		273 950	456 341	556 852	606 035	620 792
			-15	0	2	1	1
			-240/-4	1/0	35/1	18/0	12/0
			---	9/0	---	280/0	---
.70	Not		4316/1079	1893/473	-738/-184	-1984/-496	-2306/-576
			1 186	561	-190	-529	-611
	Computed		174 521	312 202	402 754	452 280	467 902
			-3	-2	0	0	1
			-39/-1	-33/-1	1/0	9/0	9/0
.60	Not		---	5729/358	---	-2043/-128	---
			2327/582	1895/474	460/115	-600/-150	-957/-239
			594	517	133	-157	-255
	Computed		118 017	217 756	289 503	331 588	345 350
			0	0	0	0	0
			-9/0	-13/0	-5/0	2/0	3/0
			---	-127/0	---	33/0	---

		1299/325	1450/362	855/214	239/60	0/0
	Not	322	376	228	64	0
.50		81 588	152 754	206 341	239 060	250 000
	Computed	0	-1	0	-1	0
		-3/0	-6/0	-4/0	-2/0	0/0
		—	3813/238	—	2042/128	—
		773/193	1046/262	878/220	600/150	478/120
	Not	190	266	228	157	126
.40		56 226	106 035	144 440	168 412	176 531
	Computed	0	0	0	0	0
		-2/0	-2/0	0/0	-2/0	-1/0
		—	0/0	—	-33/0	—
		478/120	728/182	737/184	646/162	596/149
	Not	118	183	189	168	155
.30		37 549	71 076	97 247	113 776	119 416
	Computed	0	0	-1	0	-1
		0/0	-2/0	-2/0	-1/0	-2/0
		—	1795/112	—	1797/112	—
		285/71	465/116	521/130	508/127	494/124
	Not	70	116	133	131	128
.20		23 022	43 659	59 870	70 173	73 701
	Computed	0	0	0	0	0
		-1/0	-1/0	-1/0	-1/0	-1/0
		—	-9/0	—	-15/0	—
		134/34	227/57	268/67	275/69	273/68
	Not	33	57	68	71	70
.10		10 941	20 766	28 507	33 441	35 134
	Computed	-1	0	0	0	0
		-1/0	0/0	0/0	0/0	0/0

TABLE IV

Numerical Results for Problem Ia.

y	x	.05	.10	.20	.25	.30	.35	.40	.50
		87	196	528	971	577	7 990	2 425	647
.95		1 040	2 268	6 747	12 243	25 898	54 247	72 649	80 519
		73	164	560	1 149	3 290	5 310	2 286	882
			-645	-2 308		-8 108		-999	-1 747
		Not	293	705	Not	1 285	Not	2 718	1 402
.90			4 106	11 253		29 165		55 378	63 739
		Computed	295	852	Computed	2 337	Computed	2 205	1 397
			-668	-2 078		-5 227		-3 607	-1 069
			---	347		---		9 401	---
			-967	-2 404		-3 891		-1 586	-1 422
		Not	301	634	Not	1 217	Not	-1 794	1 657
.80			5 855	13 668		24 794		36 054	40 633
		Computed	364	839	Computed	1 377	Computed	1 529	1 439
			-801	-1 820		-2 845		-2 731	-2 487
			---	3 188		---		6 627	---
			-850	-1 709		-2 146		-1 459	-1 241
		Not	254	540	Not	908	Not	1 219	1 306
.70			5 678	12 031		18 970		24 837	27 157
		Computed	316	645	Computed	936	Computed	1 083	1 118
			-662	-1 326		-1 846		-2 027	-2 033
			---	1 114		---		4 906	---
			-611	-1 120		-1 328		-1 138	-1 039
		Not	209	433	Not	673	Not	867	1 135
.60			4 713	9 492		14 021		17 442	18 735
		Computed	243	474	Computed	665	Computed	781	817
			-490	-1 943		-1 289		-1 474	-1 526
			---	1 905		---		3 372	---

		-407	-725		-868		-841	-811
	Not	164	332	Not	493	Not	615	660
.50		3 617	7 094		10 140		12 289	13 072
	Computed	117	341	Computed	474	Computed	557	586
		-351	-668		-912		-1 059	-1 107
		---	901		---		2 433	---
		-262	-367		-575		-598	-595
	Not	123	244	Not	352	Not	430	459
.40		2 631	5 090		7 153		8 552	9 049
	Computed	126	240	Computed	332	Computed	390	411
		-245	-365		-637		-745	-781
		---	999		---		1 665	---
		-163	-295		-373		-404	-410
	Not	87	169	Not	240	Not	290	309
.30		1 807	3 472		4 836		5 740	6 057
	Computed	86	162	Computed	223	Computed	263	277
		-164	-313		-429		-503	-528
		---	500		---		1 017	---
		-95	-174		-224		-248	-255
	Not	55	105	Not	150	Not	179	190
.20		1 124	2 152		2 983		3 528	3 718
	Computed	54	99	Computed	138	Computed	162	171
		-101	-193		-264		-310	-326
		---	420		---		688	---
		44	-81		-105		-118	-122
	Not	27	51	Not	72	Not	85	90
.10		538	1 028		1 422		1 679	1 768
	Computed	25	47	Computed	75	Computed	77	81
		-48	-92		-126		-148	-155

TABLE V

Numerical Results for Problem II.

y	x .50	.55	.60	.65	.70	.75	.80	.85	.90	.95
.95	(353) 117 673	(383) 115 932	(540) 110 762	(444) 102 370	(352) 91 190	(221) 77 873	(114) 63 149	(50) 47 655	(20) 31 833	(6) 15 920
.90	(629) 238 843	(716) 235 316	(892) 224 786	(945) 207 568	(729) 184 532	(435) 157 135	(220) 127 031	(103) 95 602	(48) 63 731	--- ---
.85	(727) 367 073	(920) 361 742	(1 366) 345 602	(1 604) 318 696	(1 148) 282 269	(625) 239 049	(317) 192 149	(175) 143 911	--- ---	--- ---
.80	(518) 505 886	(861) 498 988	(1 824) 477 405	(2 655) 439 645	(1 576) 386 858	(757) 324 486	(444) 258 411	--- ---	--- ---	--- ---
.75	(-24) 658 139	(362) 650 740	(1 947) 625 852	(4 658) 576 579	(1 748) 501 029	(882) 413 127	--- ---	--- ---	--- ---	--- ---
.70	(-465) 824 317	(-392) 819 038	(497) 799 133	(8 128) 743 944	(1 121) 626 304	--- ---	--- ---	--- ---	--- ---	--- ---

All values have been multiplied by 10^6 to avoid decimals. If the errors for the 5-point formula were proportional to h^2, then the numbers E_{20}, $(1/4)E_{10}$, $(1/16)E_5$ would be equal. Similarly if the errors for the 9-point formula were proportional to h^6, then F_{20}, $(1/64)F_{10}$, $(1/4096)F_5$ would be equal.

The results obtained do not appear to contradict the assumptions on the behavior of the error, although for the 9-point formula the errors for the various mesh sizes are generally so small that they are difficult to compare. The results for the 5-point case could have been anticipated from Theorem 4.1 of Walsh and Young.[16] For although the boundary values are not continuous, the discontinuities occur at mesh points and it can be shown that the error vanishes like h^2.

TABLE IV gives the corresponding results for Problem Ia except that the quantities $(1/4)E_{10}$, $(1/16)E_5$, $(1/64)F_{10}$ and $(1/4096)F_5$ are omitted. We first note that the errors are much larger than in the previous problem and that the 9-point solutions are only slightly better than the 5-point solutions. Also, since the errors have different signs for different values of h, it is clear that they are not even approximately proportional to h^2. In fact, by Theorem 4.1 of Walsh and Young,[16] the error vanishes like the first power of h as $h \to 0$, since the discontinuities do not occur at mesh points. Moreover the error is not directly proportional to h but behaves in a manner similar to the function $h \sin(1/h)$ which, although tending to zero like h, is in no sense asymptotically proportional to h.

We shall now attempt to explain the changes in sign of the error. From considerations based on the analysis given in Walsh and Young,[16] page 87, it appears that the solution of the difference equation approximates the solution of the Dirichlet problem more closely when the jumps in the function defining the boundary values occur midway between two adjacent mesh points. Thus, for the case $h = 1/20$ the difference equation solutions would have been more accurate if the jumps had occurred at (.325,1) and (.675,1) instead of at (1/3,1) and (2/3,1). This is shown by the results given in the following table where the errors at three points are given both for Problem Ia and for the modified problem. Of course the solutions of the difference equations are exactly the same for Problems Ia and Ia*.

		Error for Problem Ia with jumps at (1/3,1), (2/3,1)		*Error for Problem Ia* with jumps at (.325,1), (.625,1)*	
x	y	5-pt.	9-pt.	5-pt.	9-pt.
.3	.8	.01217	.01377	−.00167	−.00007
.3	.9	.01285	.02337	−.01226	−.00174
.4	.8	.01794	.01529	−.00336	−.00071

TABLE VI

Numerical Results for Problem III.

y	x .50	.55	.60	.65	.70
.95	(-5 170) 822 890	(-5 651) 812 665	(-5 288) 777 184	(-7 782) 695 785	(-120) 498 229
.90	(-4 026) 664 883	(-4 045) 648 4f	(-3 618) 594 187	(-859) 484 665	(7 239) 289 194
.85	(-3 273) 558 110	(-2 901) 520 205	(-1 591) 465 528	(1 405) 359 616	(4 590) 201 228
.80	(-2 134) 446 020	(-1 852) 429 855	(-970) 379 827	(642) 292 072	(1 875) 163 145
.75	(-1 011) 385 716	(-927) 372 891	(-854) 353 828	(-1 502) 266 665	(-3 905) 165 847
.70	(142) 350 862	(196) 341 979	(317) 315 592	(477) 272 748	(825) 216 760
.65	(1 112) 335 850	(1 140) 328 684	(1 191) 314 075	(1 144) 295 100	— —
.60	(1 753) 327 444	(1 722) 325 170	(1 584) 319 495	— —	— —
.55	(2 085) 325 959	(2 004) 325 464	— —	— —	— —
.50	(2 184) 325 860	— —	— —	— —	— —

We are now able to explain the changes of sign in the error. Since the exact solution of Problem Ia* is obviously larger than the exact solution of Problem Ia and since the difference equation solutions are comparatively close to the solution of Ia*, then it follows that the errors will be positive. This is actually the case. Similarly, the solution of the difference equation with $h = 1/10$ closely approximates the solution of a Dirichlet problem similar to Problem Ia but with jumps at (.35,1) and (.65,1) instead of at (1/3,1) and (2/3,1). Following this reasoning the sign of the errors for all three mesh sizes is explained.

Rather than modify the boundary conditions for the differential equation near a jump, one should be able to obtain improved accuracy by modifying the boundary value at nearby mesh points for use with the difference equation. For instance, with Problem Ia and with $h = 1/20$ it seems reasonable to replace the value unity at (.35,1) and (.65,1) by 2/3, which is the average of the boundary values in the intervals (.325,1) to (.375,1) and (.625,1) to (.675,1) respectively. This modification would certainly introduce corrections with the proper sign. It is believed that they would be of approximately the right size but this remains to be demonstrated.

TABLE VII

Numerical Results for Problem IV.

y	x	.05	.10	.20	.30	.40	.45	.50	.60	.70	.80	.90	.95
.95		(-13)	(-7 447)	(-2 607)	(-1 001)	(-578)	(-514)	(-503)	(-608)	(-1 041)	(-2 639)	(-7 464)	(-22)
		495 754	696 531	826 576	867 703	882 883	885 634	886 110	880 792	864 380	823 322	694 553	494 719
		---	(-357)	(-7 939)	(-5 781)	(-4 404)	---	(-4 149)	(-4 531)	(-5 918)	(-8 033)	(-403)	---
.90		(7 393)	(-54)	(-2 429)	(-1 455)	(-994)	(-921)	(-912)	(-1 056)	(-1 533)	(-2 489)	(-85)	(7 378)
		286 486	483 016	670 084	741 110	768 542	773 488	774 223	764 134	734 132	663 265	478 878	284 321
		---	(-208)	(-186)	(-554)	(-687)	---	(-677)	(-541)	(-337)	(9)	(-96)	---
		---	(6 498)	(-1 421)	(-4 623)	(-5 601)	---	(-6 212)	(-5 789)	(-4 747)	(-1 475)	(6 469)	---
.80		(2 495)	(2 210)	(-213)	(-1 038)	(-1 197)	(-1 256)	(-1 311)	(-1 323)	(-1 173)	(-308)	(2 161)	(2 470)
		139 475	261 993	432 006	519 690	556 740	562 873	562 768	545 911	502 768	415 714	252 197	134 362
		---	(-643)	(-845)	(-122)	(-1 511)	---	(-1 491)	(-1 126)	(-686)	(-391)	(-394)	---
		---	(2 427)	(297)	(-3 410)	(-5 412)	---	(-7 434)	(-5 349)	(-2 755)	(598)	(3 623)	---
.70		(816)	(1 093)	(379)	(-449)	(-946)	(-1 192)	(-1 392)	(-1 123)	(-552)	(314)	(1 049)	(791)
		81 491	157 199	276 342	346 345	374 209	376 147	372 322	351 658	312 265	244 440	138 328	71 680
		---	(-648)	(-1 312)	(-2 031)	(-2 673)	---	(-2 718)	(-1 755)	(-936)	(-488)	(-233)	---
		---	(1 153)	(-564)	(-3 397)	(-5 694)	---	(-10 184)	(-3 757)	(-1 272)	(511)	(1 429)	---
.60		(288)	(433)	(183)	(-365)	(-696)	(-976)	(-1 960)	(-649)	(-168)	(227)	(396)	(260)
		49 716	96 592	172 609	215 714	221 649	211 905	198 074	174 474	149 941	113 941	62 672	32 171
		---	(-751)	(-1 629)	(-2 808)	(-4 505)	---	(-5 171)	(-2 033)	(-775)	(-338)	(-133)	---
		(165)	(246)	(35)	(-515)	(-952)	(-660)	(-2 847)	(-224)	(-53)	(120)	(184)	(117)
.55		38 758	75 327	134 276	164 779	157 649	136 855	109 573	87 604	74 282	56 027	30 631	15 694
		---	(-77)	(-1 669)	(-5 097)	(-11 059)	---	---	---	---	---	---	---
		(89)	(126)	(-66)	(-633)	(-1 963)	(-3 323)	---	---	---	---	---	---
.50		30 024	58 295	103 149	123 161	105 873	73 749	---	---	---	---	---	---
		---	(-750)	(-1 657)	(-3 050)	(-6 004)	---	---	---	---	---	---	---
		---	(-274)	(-1 075)	(-2 124)	(-1 938)	---	---	---	---	---	---	---
.40		(29)	(38)	(-44)	(-197)	(-47)	(215)	---	---	---	---	---	---
		17 545	33 919	58 668	65 772	47 175	26 435	---	---	---	---	---	---
		---	(-617)	(-1 291)	(-2 033)	(-2 472)	---	---	---	---	---	---	---
		---	(-104)	(-302)	(-385)	(-63)	---	---	---	---	---	---	---
.30		(24)	(44)	(65)	(103)	(178)	(143)	---	---	---	---	---	---
		9 810	18 871	31 902	34 080	22 550	12 131	---	---	---	---	---	---
		---	(-425)	(-824)	(-1 094)	(-917)	---	---	---	---	---	---	---
		---	(28)	(54)	(123)	(189)	---	---	---	---	---	---	---
.20		(25)	(49)	(95)	(135)	(127)	(78)	---	---	---	---	---	---
		5 112	9 796	16 292	16 922	10 828	5 754	---	---	---	---	---	---
		---	(-249)	(-453)	(-535)	(-384)	---	---	---	---	---	---	---
		---	(46)	(94)	(138)	(126)	---	---	---	---	---	---	---
.10		(16)	(31)	(60)	(78)	(61)	(35)	---	---	---	---	---	---
		2 164	4 138	6 819	6 986	4 409	2 332	---	---	---	---	---	---
		---	(-12)	(-196)	(-217)	(-147)	---	---	---	---	---	---	---

In TABLES V, VI, and VII the results for Problems II, III, IV, respectively, are given. For each point we give

$$(u_{1/10} - v_{1/20}) \text{ (when available)}$$

$$(u_{1/20} - v_{1/20})$$

$$v_{1/20}$$

$$(v_{1/10} - v_{1/20}) \text{ (when available).}$$

(All values are multiplied by 10^6.)

Since in each case the exact solution of the Dirichlet problem is not available and since only one or at most two different mesh sizes could conveniently be used, we cannot conclude much about the behavior of the error as a function of h. However, if we assume that the differences between the solution of a given difference equation and $v_{1/20}$ give a fair indication of the accuracy, it appears that the results of using the 5-point difference equation for Problems II, III, IV are about as accurate as for Problem I. Also the errors occurring in the presence of a corner with an interior angle of 270° seem to be of about the same size as those observed near a discontinuity which occurs, as in Problem I, at a mesh point. Near such corners the derivatives of the solution of the Dirichlet problem are not bounded even when the boundary values and the function itself are continuous there. It remains to be determined whether, as in the case of the discontinuity at a mesh point, the error is approximately proportional to h^2.

It is evident from this discussion that much work, both theoretical and experimental, remains to be done before the accuracy question will be resolved.

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